

Lab Report 4

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ABSTRACT

In this lab report, I have processed images of The Orion Nebula (M42), The Crab Nebula (M1), and Jupiter. We captured bias images, darks, and flats at different wavelengths and exposure times to reflect our observations. In addition, we captured three images of these objects at four different wavelengths. The purpose of this lab is to combine our raw images into a representative RGB image that shows what planets and nebulae look like. The RGB images of the Orion and Crab Nebula show how gas clouds are structured and visible through different wavelengths. The images of Jupiter show all four Jovian moons in orbit, as well as a detailed image of the Jupiter's cloudy atmosphere.

1. INTRODUCTION

The Orion Nebula is a nebula with an apparent magnitude of +4.0 and is 1.5 thousand light years away in the Larger Orion Molecular Cloud Complex. The nebula itself is about 24 light years across. M1, or the Crab Nebula, is a supernova remnant with an apparent magnitude of +8.4. It is nearly 6,500 light years away and in the constellation Taurus [NASA \(2017\)](#). Jupiter is the fifth planet from the sun and the largest in the solar system by both size and mass. It is also one of the brightest objects in the sky after the Moon and Venus. Jupiter is a gas giant and has an atmospheric bands containing dense cloud layers that we were able to capture with the telescope [Astronomy Index \(2021\)](#). To combine an RGB image, we used python's built in libraries like Pyplot and NumPy. Using these libraries, I was able to easily modify the data per wavelength and combine them in a three dimensional array. Exponential scaling was also utilized for the nebulae. This consists of putting every pixel to some power to exaggerate the counts.

2. METHODS

On the first night (2/17/2025) we used infrared, red, visible, and blue wavelengths to take images on UF's 14" telescope. We took darks at 0.4 seconds, one second, ten seconds, twenty seconds, thirty seconds, sixty seconds, and one hundred seconds. This time, we had the sky to take flats and we did at 0.2s for blue (36,000 counts), 0.3s for red (37,500 counts), 0.3s for visible (40,000 counts), 1s for infrared (36,000 counts), and 75ms for clear (37,000 counts). Then we calibrated our telescope using Polaris and Sirius at basis stars. After calibration we began by capturing images of M42 at a sixty second exposure time. Next we captured M1 also at a sixty second exposure time. On the second night (3/3/2025) we used infrared, red, visible, and blue wavelengths to take images on UF's 11" telescope. We took darks at 0.001 seconds, 0.1 seconds, one second, ten seconds, thirty seconds, sixty seconds, and one hundred seconds. For our flats we used reflected light on the wall to capture them. This was done at 0.5s for clear (36,000 counts), 8s for blue (47,000 counts), 2s for visible (38,000 counts), 1s for red (41,000 counts) and 15s for Infrared (34,000 counts). We calibrated our telescope using sirius and capella as basis stars. We then took images of Mars at 7:23 P.M. at 0.1s exposure time. We then took images of Mars at 8:00 P.M. at 1s exposure times. When combining and processing data I used Anaconda's Jupyter Notebooks as well as the Matplotlib, Astropy, Pylab, Glob, and Numpy Python libraries. I created a master bias and dark frame to subtract from the images and a flat image to divide the images. After that I used Pylab to plot the images. I then used Numpy array features in order to modify the scaling of the raw images and combine them into a colorful and representative image. Part of this scaling was raising each pixel to a certain power in order to enhance some of the cloudy structures captured in our images. M42 was raised to the 0.7 power. My images were also scaled linearly in each wavelength as well as a collective scaling value. These varied for each image.

3. DATA

3.1. *The Orion Nebula*

The Orion Nebula is the first image we captured in four different wavelengths and has the most visible structure out of the planets and nebulae we have captured. You can very clearly see the cloud structures in the image, and the strength of green in the center of the nebula. This indicates that it is very bright in the visible spectrum towards the center, and red on the edges. The image below shows M42 in red, visible, and blue light at sixty second exposure time.



Figure 1. The Orion Nebula in red, visible, and blue light

3.2. *The Crab Nebula*

The Crab Nebula is the next we captured. In this nebula too you can very clearly see some cloud structures in the images we captured. The strong concentration of red in the gas cloud emphasized how the nebula peaks in infrared wavelengths. The image below shows M1 in infrared, visible, and blue light at an exposure time of sixty seconds.

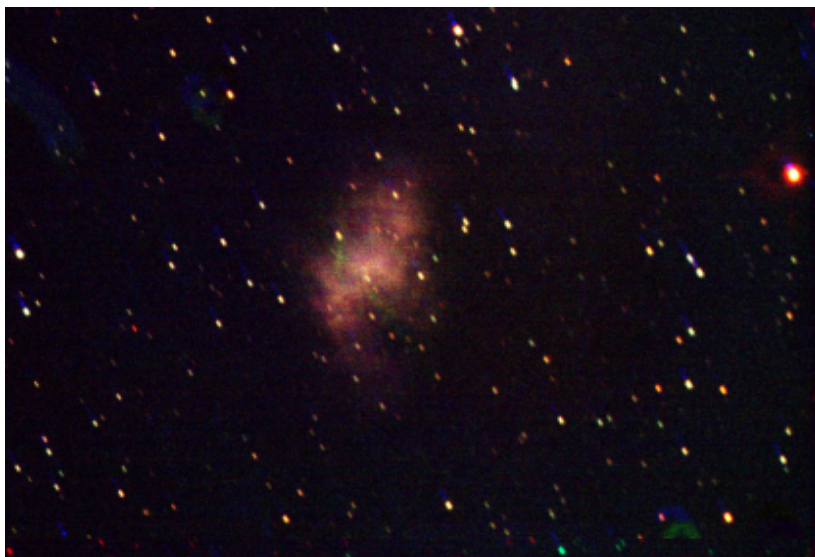


Figure 2. M1 in infrared, visible, and blue light

3.3. *Jupiter*

On the next night we imaged Jupiter at 0.1 second exposure time and were able to see not only the planet but it's four largest moons. In the first image, the moons are visible in order from top to bottom: Callisto, Io, Europa, Ganymede. After scaling the image to 0.0025 times it's original counts value, I was able to clearly make out the details of the planet as shown in the second image. The images below shows Jupiter in red, visible, and blue light.

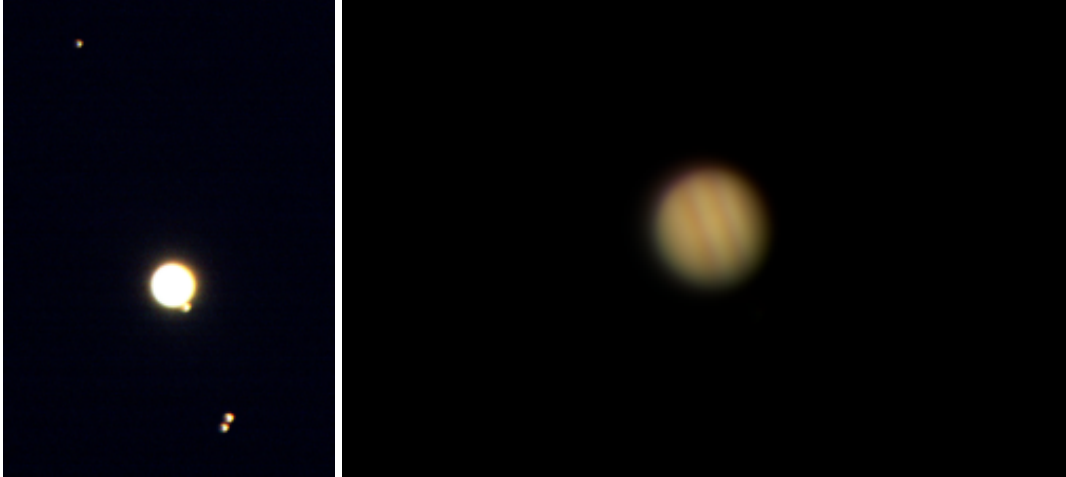


Figure 3. Jupiter and its moons in red, visible, and blue light

4. QUESTIONS

Describe the images and the objects within them. Describe the colors of the object(s). Is every part of the image the same color? Do you have thoughts on astrophysical reasons why different parts might be different colors?

For M42 the center of the nebula is more green while the rims are more reddish. I believe this is because visible light emits at higher temperatures and it would make sense that it would be hotter at the center of the nebula. Although blue emits at an even higher temperature, it is exaggerated more than red or visible light so it would still make sense that the visible light corresponds to where the heat of the nebula is. This is also somewhat true with the Crab Nebula. In the image there are green bands close to the center which I believe are there due to parts of the nebula that are hotter or have more energy, and are therefor radiating more energized light. Jupiter also has variation in color, however I think that has to do with the bands of clouds on Jupiter consisting of different compounds, and therefor reflecting different types of light.

Measure the flux from each object in counts. Explain what part you have chosen to measure and why. What are the signal to noise ratios of your measurements?

I calculated the signal to noise ratio by only including the pixels I wanted (removing the background noise) and summing all of their counts. I then divided by the exposure time to get photons/second. To get the SNR I took the square root of the signal. The results are recorded in the chart below.

	Counts (photons/s)	SNR
Jupiter Red	1.4E+09	3.8E+04
Jupiter Visible	1.0E+09	3.2E+04
Jupiter Blue	5.1E+08	2.2E+04
M42 Infrared	6.6E+06	2.6E+03
M42 Visible	3.0E+07	5.4E+04
M42 Blue	2.6E+07	5.1E+04
M1 Infrared	2.0E+05	4.5E+02
M1 Visible	1.2E+04	3.4E+02
M1 Blue	9.8E+04	3.1E+02

Figure 4. Table of fluxes and signal to noise ratios

5. ERROR ANALYSIS

There are many aspects of this experiment that could contribute to error. For our last night we captured the flats too late, forcing us to use the wall of the observatory and the projector. This could have caused errors with the flats since they were not aimed at the sky. During night two, while capturing images of M42 we used the OIII filter, so we ended up using infrared images from our tests instead of our intentional captures. Finally, we did not use the finder charts to find the objects we were looking for during our second night. This led to some difficulty finding our objects. The signal calculated could be off since it might have included the brightness from nearby stars or gas that is not apart of the nebulae or planets themselves. I tried to remove as much background noise as I could however it was difficult to remove anything brighter than the objects themselves, leading to some potentially misleading results.

6. CONCLUSIONS

In this lab I successfully combined images of the Orion Nebula, the Crab Nebula, and Jupiter. These images are captured in different wavelengths and scaled to be visually appealing and representative of their contents. The nebulae are green in the middle and reddish on the outside, indicating it is hotter in the interior of the nebulae. I believe that I completed my goals for this lab and am very happy with the data from lab 3, and the outcome of the images in this lab. In the future I would like to experiment with different exposure times for the objects to see if I can get better results, since we only did what we thought the best one would be. I feel that this lab is good and I do not have any improvements for it in the future.

REFERENCES

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